Conjugate Heat Transfer Analysis and Design Optimisation of a Microchannel Cold Plate for High-Power AI Processors

Chanduri Sreekrishna Abhay
B.Tech. Aeronautical Engineering, Jain (Deemed-to-be University), Bengaluru
https://github.com/KrishnaChanduri06/microchannel-cold-plate-cht

October 2026

Abstract

A three-dimensional conjugate heat transfer (CHT) model of a single-phase copper microchannel heat sink was built in Ansys Fluent 2026 R1, scripted in Python (PyFluent) with a structured hexahedral mesh generator, and validated against the experiments of Qu and Mudawar (2002). Across six operating points ($Re = 226$–1454, $q''_{\text{eff}} = 100$ and $200\,\text{W/cm}^2$) the CFD predicts the thermocouple temperatures with a mean error of 5.4 %, matching the original authors' CFD (5.6 %), and closes the energy balance within 1.5 %. A three-mesh Grid Convergence Index study gives discretisation uncertainties below 4.6 %, and a no-heating case Richardson-extrapolates to within 0.1 % of the analytical laminar developing-flow pressure drop. A 1-D resistance-network model with a fully developed three-wall Nusselt number (computed here as 5.30) over-predicts temperatures by up to 62 %; the error is governed almost entirely by the ratio of thermal entry length to channel length. The validated model was then used for an automated 36-case design study of a product-like cold plate (30 °C water, $100\,\text{W/cm}^2$): narrow channels with thin fins dominate the thermal-resistance/pumping-power Pareto front, and silicon is 15–35 % worse than copper at identical geometry. A multi-channel Ansys Icepak model confirms that the base temperature is uniform across the plate width, supporting the unit-cell assumption.

1 Introduction

Data-centre AI accelerators now dissipate around 1 kW per package at average die fluxes near $100\,\text{W/cm}^2$, beyond the reach of air cooling. Liquid cold plates with microchannels are the established solution, and channels embedded in the die are under active research. This work builds and validates a CHT model of a classic benchmark microchannel heat sink [1], quantifies where simple correlation-based sizing fails, and uses the validated model to explore cold-plate design trade-offs.

2 Benchmark and models

2.1 Benchmark

Qu and Mudawar [1] tested an oxygen-free copper heat sink with 21 channels of $231\,\mu\text{m} \times 713\,\mu\text{m}$ cross-section and 44.764 mm length, closed by a polycarbonate cover, cooled by 15 °C water at $Re = 139$–1672, and heated with $q''_{\text{eff}} = 100$ and $200\,\text{W/cm}^2$ over a $1.0 \times 4.48\,\text{cm}^2$ planform. Four thermocouples sit 3.175 mm above the heated surface. Figures 4–7 of the paper were digitised automatically (marker detection and curve tracing with OpenCV, calibrated against the axis ticks); the three figures that share flow rates agree in Re to 0.2 %.

2.2 1-D analytical model

Friction uses the Shah & London fully developed $fRe = 17.2$ ($\alpha = 0.324$) plus the Hagenbach entrance factor $K(\infty) = 1.18$ [2]. Because the cover is almost adiabatic only three walls are heated; the corresponding fully developed H1 Nusselt number was computed with a 2-D finite-volume solver, giving $Nu_3 = 5.30$ (the solver reproduces the four-wall values 3.608, 4.124, 5.333 to ± 0.002). Fins have efficiency $\eta_f = \tanh(mH)/(mH) \approx 0.97$, and conduction, convection and caloric resistances are combined in series. Re uses inlet properties as in the paper; ΔP and h use mean-bulk properties.

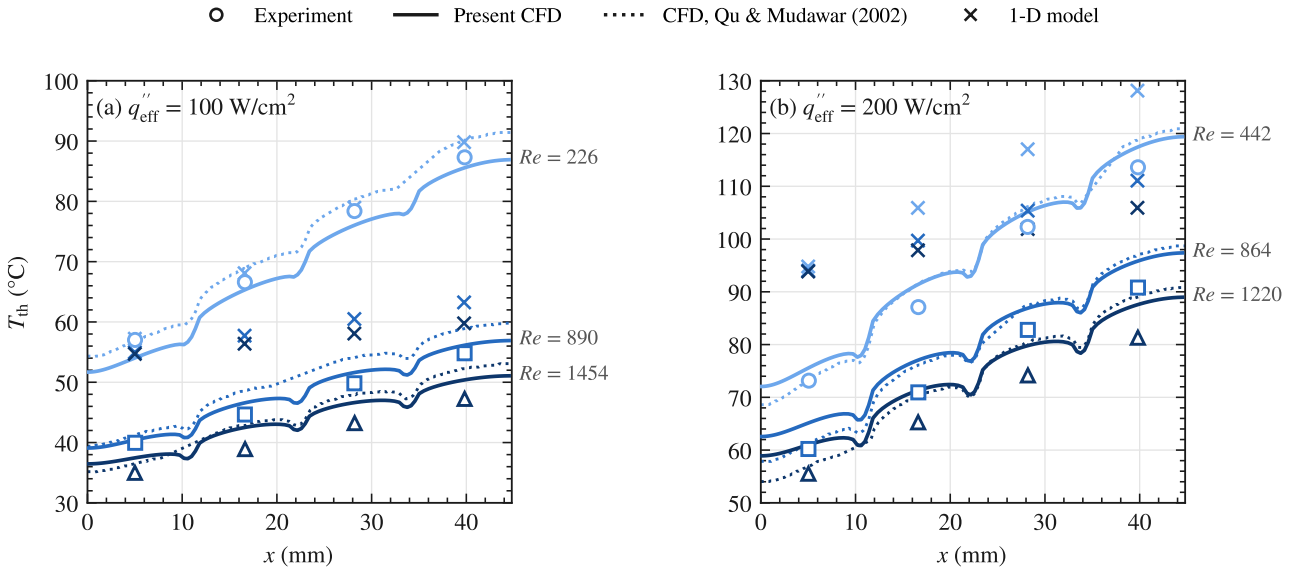


Figure 1: Temperature along the thermocouple line: experiment (open markers), present CFD (solid), CFD of Qu and Mudawar (dotted) and the 1-D model at the thermocouple positions (×). The dips mark the three air slots.

2.3 CFD model

The unit cell (one channel, two half fins, copper base, cover, and the three 1.6 mm air slots of the test section, Fig. 5) is meshed with a structured hexahedral mesh generator that writes Fluent's native `.msh` format (orthogonal quality 1.0; baseline 89 760 cells, clustered at walls and inlet). The flow is steady and laminar; water $\mu(T)$, $k(T)$ and $c_p(T)$ are polynomial fits to CoolProp (max error 0.06 %); density is constant. Boundary conditions follow the paper: uniform inlet velocity at 15 °C, zero-gauge outlet, a uniform heat flux on the copper part of the base ($q''_{\mathrm{bot}} = P_W/(N A_{\mathrm{bot}}) = 114.3\,\mathrm{W/cm^2}$ at the nominal 100), $h = 10\,\mathrm{W/m^2K}$ on the cover top, and symmetry sides. The pressure-based coupled solver with second-order schemes is iterated until the thermocouple temperatures change by less than 0.005 K and ΔP by less than 0.01 % per 100 iterations (about 500 iterations, 90 s per case on 4 cores).

3 Verification

Table 1: Grid Convergence Index [3] at 100 W/cm^2, $Re = 890$ (meshes of 29 820 / 89 760 / 297 891 cells, $r \approx 1.5$).

Quantity	coarse	medium	fine	extrapolated	p	$\mathrm{GCI}_{\mathrm{fine}}$
ΔP_{ch} [bar]	0.325	0.334	0.338	0.341	2.3	1.0 %
$T_{\mathrm{heater,avg}} - T_{\mathrm{in}}$ [K]	42.42	42.47	42.71	42.77	3.8	0.2 %
$T_{\mathrm{heater,max}} - T_{\mathrm{in}}$ [K]	50.07	50.66	51.12	52.12	0.9	2.4 %
$T_{\mathrm{TC3}} - T_{\mathrm{in}}$ [K]	36.16	36.61	36.99	38.37	0.6	4.6 %
ΔP_{ch}, no heating [bar]	0.415	0.442	0.455	**0.467**	1.9	3.3 %

All quantities are mesh-converged to within 5 % on the baseline mesh (Table 1). For the no-heating case the laminar developing-flow pressure drop is known analytically, $[4fL/D_h + K(\infty)]\,\rho u^2/2 = 0.4678$ bar; the Richardson-extrapolated CFD value of 0.4672 bar agrees within 0.1 %, with observed order 1.9, which verifies the solver set-up, boundary conditions and mesh writer.

4 Validation

The present CFD reproduces the measured temperature distribution as closely as the published CFD and follows it point by point, including the slot dips (Fig. 1, Table 2). The residual 5–12 % warm bias grows with Re in both CFD studies; plausible causes are thermocouple contact and lead conduction, which normally make thermocouples read low, the unspecified slot height, and the less-than-1 % heat loss. The measured ΔP (Fig. 2) includes plenum losses that are not geometrically specified; adding one velocity head of loss to the CFD channel

Table 2: Mean absolute error against the experiment at the six validation points.

Quantity	present CFD	Qu & Mudawar CFD	1-D model
Thermocouple temperature, Fig. 7	**5.4 %**	5.6 %	28.5 %
Thermocouple rise $T_{tc} - T_{in}$, Fig. 6	**7.2 %**	–	40.5 %
Water temperature rise, Fig. 5	1.5 %	–	1.5 %
Channel pressure drop, Fig. 4b	10.5 %	18.8 %†	35 %

† includes their plenum contraction/expansion losses.

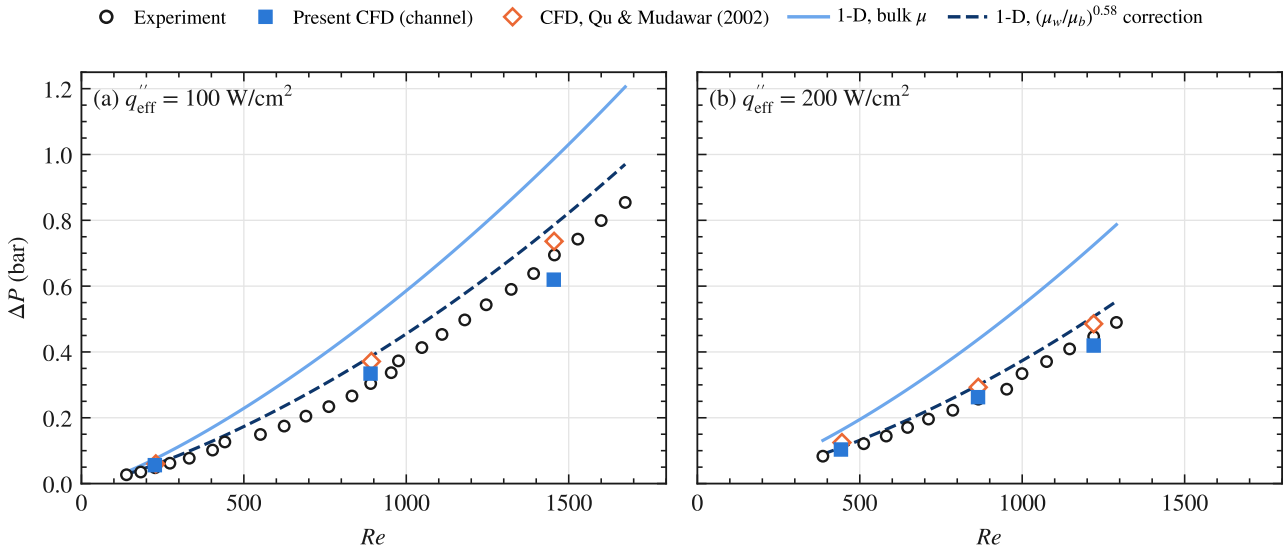


Figure 2: Pressure drop: experiment, present CFD (channel only), CFD of Qu and Mudawar (including plenum losses) and the 1-D model with bulk viscosity and with the wall-viscosity correction.

value raises the error to 24 %, which suggests strong pressure recovery in the outlet plenum. Heating reduces ΔP markedly (0.44 bar cold vs 0.33 bar heated at $Re = 890$) because the hot walls lower the near-wall viscosity, which is the main reason bulk-property correlations over-predict it; a $(\mu_w/\mu_b)^{0.58}$ correction [4] cuts the 1-D over-prediction from 56–65 % to 13–21 %.

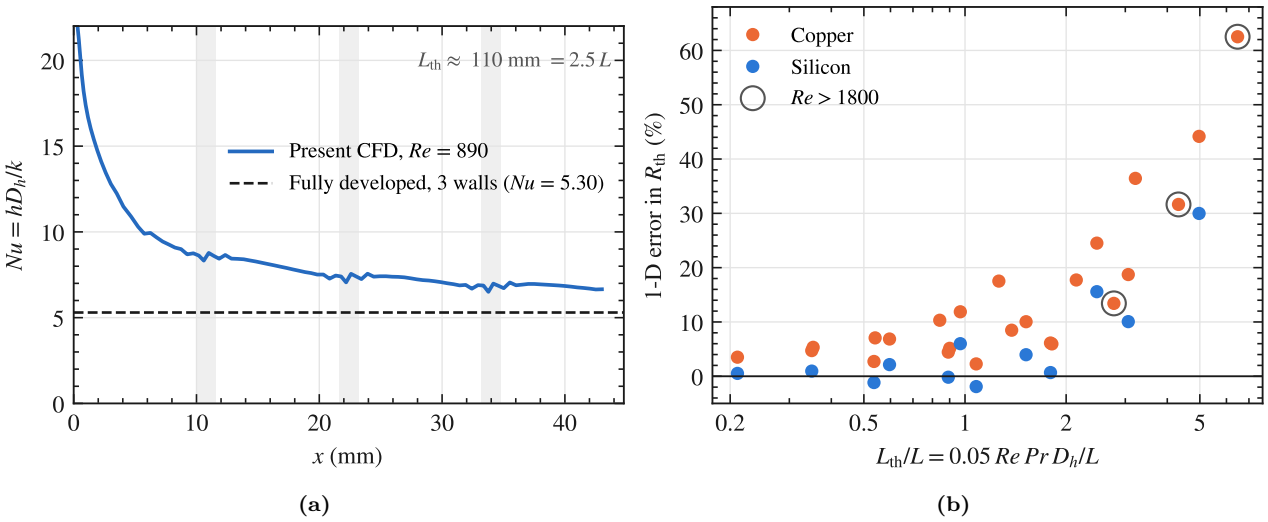
5 Why the 1-D model fails: thermal development

At $Re = 890$ the thermal entry length $L_{th} = 0.05\, Re\, Pr\, D_h \approx 110$ mm is 2.5 times the channel length, so the boundary layer never becomes fully developed: the CFD local Nusselt number falls from above 20 at the inlet to about 7 at the outlet, with a channel mean of 9.0, which is 1.7 times the fully developed 5.30 (Fig. 3a). The fully developed 1-D model therefore over-predicts temperatures most at high Re and near the inlet (TC1). Across the design study (Fig. 3b) the 1-D error is 0–7 % for $L_{th}/L < 1$ and rises to 30–62 % for $L_{th}/L > 3$. Fully developed correlations are adequate for sizing while the thermal entry length is shorter than the channel, and beyond that CFD or developing-flow correlations are needed.

6 Design study

A product-like plate was studied: a 1 mm base, a 1 mm lid of the same metal (so all four channel walls carry heat), no slots, 30 °C inlet water and a uniform 100 W/cm^2 over the 1×4.48 cm^2 footprint. A PyFluent driver swept channel width (100–350 µm), fin width (100, 236 µm), total flow (200–1000 ml/min) and solid (copper, silicon), 36 cases with energy balances of 0.999–1.000. The figure of merit is $R_{th} = (T_{base,max} - T_{in})/Q$.

- **Thin fins and narrow channels form the whole Pareto front** (Fig. 4): a smaller D_h raises h and the finer pitch adds wetted area, cutting the dominant convective resistance, at the cost of ΔP. The best copper design (100 µm/100 µm, 1000 ml/min) reaches 0.034 K/W, a 45 °C base at 448 W, for 3.3 W of pumping power. A design near the knee (150 µm/100 µm, 500 ml/min) needs ten times less pumping power for 1.7 times the resistance.



(a) **(b)**

Figure 3: (a) Local Nusselt number from CFD along the channel at $Re = 890$ (shaded: air slots) against the fully developed three-wall value. (b) 1-D thermal-resistance error vs CFD across the design study, against the thermal entry length ratio.

- **Silicon is 15–35 % worse than copper** at identical geometry (η_f 0.68–0.89 vs 0.84–0.96). Silicon microchannels pay off only when they are etched into the die [5], eliminating the thermal interface material, whose resistance typically exceeds this penalty.
- **At 200 W/cm²** R_{th} changes by less than 1.2 % (thermally linear) while ΔP falls 9–14 % (lower viscosity). Copper Pareto designs keep the base at or below 81 °C; the silicon knee design reaches 95 °C.
- Four high-flow cases with thick fins or wide channels have inlet $Re = 1860$–2300, near laminar transition; none is Pareto-optimal.

7 Multi-channel Icepak model

To test the unit-cell assumption the heat sink was also modelled in Ansys Icepak (AEDT 2025 R2, scripted with PyAEDT): channels, fins, thicker edge walls, slots and cover, with constant water properties at the mean bulk temperature. Resolving the flow in the 0.231 mm channels beside thin fins required cells of about 0.03 mm across and through the channels; coarser meshes closed the channels. At that resolution the full 21-channel plate needs 3.1 M cells, above the Student licence limit, so a 7-channel strip with the same pitch and edge walls was solved instead. The base temperature under the outermost channel equals the centre value to 0.01 K (72.66 °C both), which confirms that copper spreading makes the unit-cell idealisation sound for the solid, although the edge channel's water runs warmer because it also absorbs the heat of the thicker side wall. Absolute temperatures are 7–15 K above the validated Fluent model, worst near the inlet: Icepak's uniform cut-cell mesh (0.25 mm axial cells, no wall clustering) under-resolves the thin entrance boundary layer that dominates heat transfer there. For microchannels a structured, wall-clustered mesh is considerably more effective than a larger uniform mesh.

8 Conclusions

1. A scripted Fluent CHT workflow (structured hexahedral mesher, PyFluent driver) reproduces the Qu & Mudawar experiments with 5.4 % mean temperature error, equal to the original CFD, and verifies against analytical theory to 0.1 %.
2. Fully developed 1-D models over-predict microchannel temperatures by up to 62 % because the flow is thermally developing; the error scales with L_{th}/L, which gives a simple validity criterion.
3. For a 100 W/cm² AI-class cold plate, narrow channels with thin fins dominate the R_{th}–pumping-power trade-off; copper beats silicon by 15–35 % unless the silicon removes the interface layer.

Limitations and next steps: the chip stack and thermal interface are not modelled, the inlet and outlet plenums are excluded, and only straight channels were studied. Natural extensions are manifold microchannels, non-uniform hotspot heating, and a physical test rig.

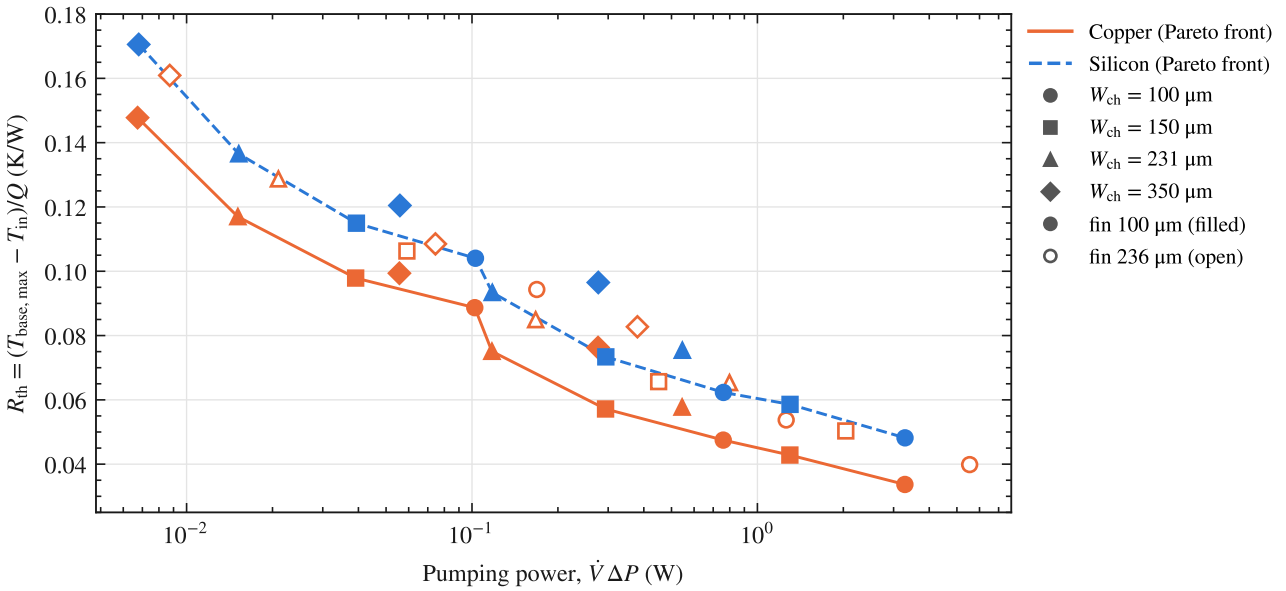


Figure 4: Thermal resistance against channel pumping power for the design study at 100 W/cm^2 (448 W, 30 °C water). Marker shape: channel width; filled: 100 µm fins, open: 236 µm fins; lines: Pareto fronts.

References

[1] W. Qu, I. Mudawar, Experimental and numerical study of pressure drop and heat transfer in a single-phase micro-channel heat sink, *Int. J. Heat Mass Transfer* 45 (2002) 2549–2565.

[2] R. K. Shah, A. L. London, *Laminar Flow Forced Convection in Ducts*, Academic Press, 1978.

[3] I. B. Celik et al., Procedure for estimation and reporting of uncertainty due to discretization in CFD applications, *J. Fluids Eng.* 130 (2008) 078001.

[4] S. Kakaç, Y. Yener, *Convective Heat Transfer*, 2nd ed., CRC Press, 1994, Eq. (9.34b).

[5] R. van Erp, R. Soleimanzadeh, L. Nela, G. Kampitsis, E. Matioli, Co-designing electronics with microfluidics for more sustainable cooling, *Nature* 585 (2020) 211–216.

A Field results

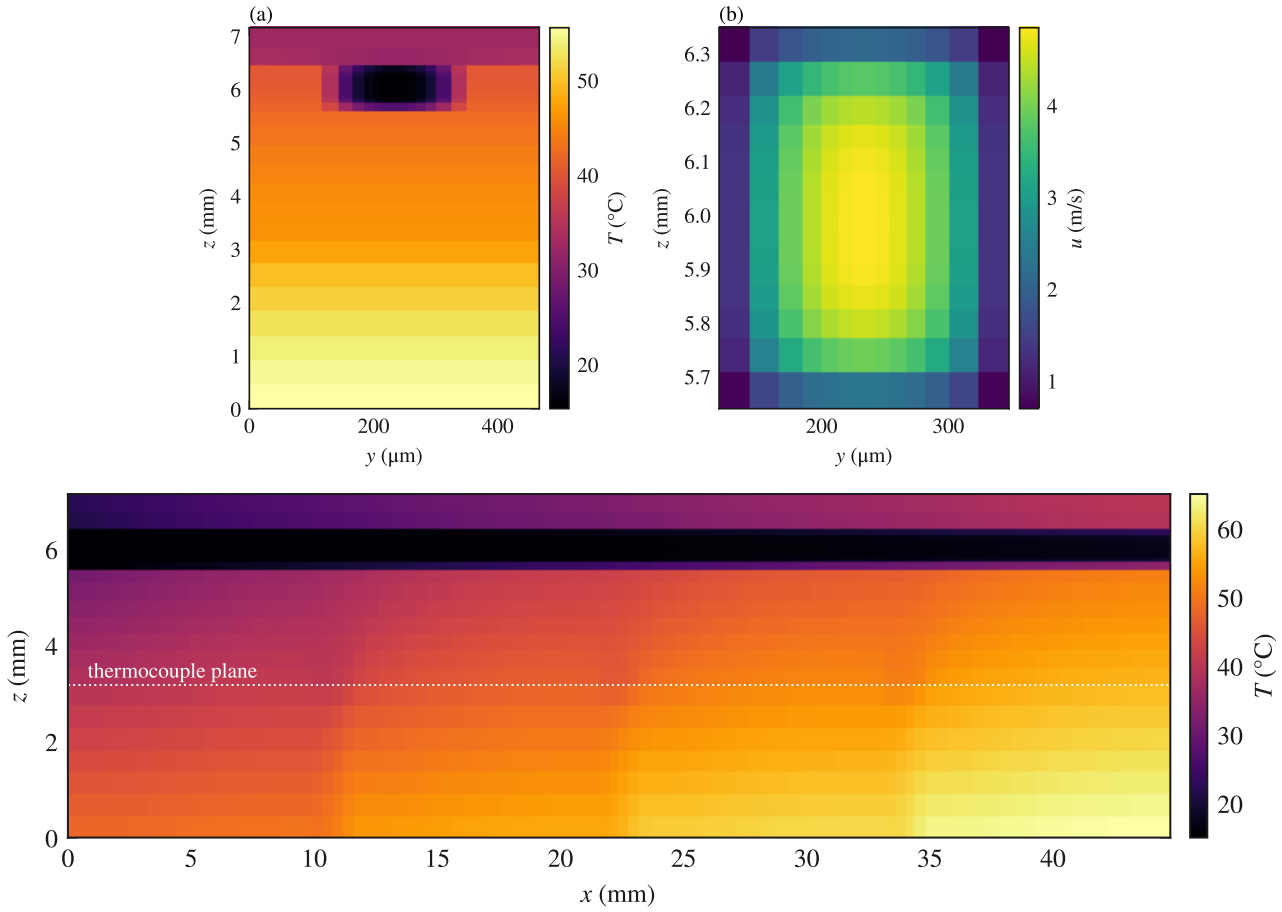


Figure 5: CFD solution at 100 W/cm^2, $Re = 890$. Top: cross-section at mid-length, (a) temperature and (b) axial velocity. Bottom: temperature on the channel mid-plane; the steps in the base are at the three air slots.